

OSHA Severe Injury Report

Capstone Project — Data Analytics

January 2015 – October 2025

Total Records: 105,317

Source: OSHA Severe Injury Dashboard | osha.gov/severe-injury-reports

1. Dataset Overview

Each row in the OSHA Severe Injury Report dataset represents a single workplace injury incident reported to OSHA between January 2015 and October 2025. Each record captures details about the employer, location, nature of the injury, body part affected, and outcome including whether the incident resulted in hospitalization, amputation, or loss of eye.

The dataset was obtained directly from the OSHA Severe Injury Dashboard at osha.gov. Column definitions were referenced from the OSHA reporting guidelines which define hospitalization, amputation, and loss of eye as the three primary severe injury outcome classifications required to be reported to OSHA.

2. Data Dictionary

2.1 NAICS Structure Reference

Reference: <https://www.census.gov/naics/?48967>

2.2 OSHA Severe Injury Dataset — Column Definitions

Column	Definition
ID	A unique OSHA case identifier assigned to each severe injury report. Used to distinguish individual incidents.
UPA	Unique Processing Activity ID. An internal OSHA submission identifier used for tracking employer-submitted reports.
EventDate	The date the injury occurred. This is the primary field for time-series analysis.
Employer	Name of the employer where the incident occurred.
Address1 / Address2	Street address of the incident location. Address2 is often blank or used for suite/building numbers.
City / State / Zip	Geographic location of the incident.
Latitude / Longitude	Geospatial coordinates of the incident location.
Primary NAICS	The employer's North American Industry Classification System code.
Hospitalized	Count of employees hospitalized. OSHA requires reporting of hospitalizations within 24 hours.
Amputation	Count of amputations that occurred in the incident.
Loss of Eye	Count loss of eye reports that occurred in the incident.
Inspection	OSHA inspection ID if an inspection was opened as a result of the incident. Blank = No inspection initiated.

Final Narrative	Employer-submitted description of what happened.
Nature / NatureTitle	Numeric OSHA code and plain-language description of the injury type (e.g., Fractures, Burns, Amputations).
Part of Body / Part of Body Title	Numeric OSHA code and plain-language description of the body part affected (e.g., Finger(s), Lower leg, Elbow).
Event / EventTitle	Numeric OSHA code and plain-language description of how the injury occurred.
Source / SourceTitle	Numeric OSHA code and plain-language description of the primary source of the injury.
Secondary Source / Title	Numeric OSHA code and plain-language description of any secondary contributing source.
FederalState	1 = Federal OSHA jurisdiction 2 = State-plan OSHA jurisdiction.

3. Data Cleaning

3.1 Duplicate Check — ID Column

A duplicate check was performed on the ID column across 105,318 records. Only one true duplicate was identified, indicating the dataset is largely clean and reliable. The duplicate record was removed prior to analysis, resulting in a final dataset of 105,317 records.

3.2 EventDate Column

The EventDate column was initially stored as text rather than a recognized date format, which prevented accurate date-based filtering and analysis. To resolve this, the Text to Columns feature in Excel was used to convert the column from a text string to a proper date format using the MDY (Month/Day/Year) setting. This ensured that all date values were recognized correctly and could be used reliably in formulas, filters, and visualizations going forward.

3.3 Employer Column

The Employer column contained 3,131 blank records out of 105,317, representing approximately 3% of the dataset. These records were retained as the remaining fields contained valid injury data useful for analysis. One employer name contained the string #NAME? and was retained in the dataset. Employer names were standardized to proper case and trimmed of leading and trailing spaces. Due to the nature of manual data entry in the OSHA reporting system, variations in employer name formatting such as abbreviations, inclusion or exclusion of legal suffixes like Inc or LLC, and shortened names may still exist in the dataset. These variations were noted as a limitation of the data source rather than a data cleaning error.

3.4 State Column

The State column was standardized to uppercase and trimmed of leading and trailing spaces. The column returned 56 unique values. Upon investigation all values were confirmed as valid; representing all 50 US states, the District of Columbia, and five US territories including Puerto Rico, Guam, American Samoa, the Northern Mariana Islands, and the Virgin Islands. OSHA has federal jurisdiction over all US territories which accounts for their presence in the dataset.

3.5 Zip Code Column

The Zip column was found to contain zip codes missing leading zeros, which occurred because Excel stored the values as numbers rather than text. The TEXT function was used to standardize all zip codes to

a five-digit format, restoring any leading zeros that had been dropped. The column was then reformatted as text to prevent this issue from recurring.

3.6 City Column

The City column contained 17 blank records out of 105,317, representing less than 0.02% of the dataset. These records were retained as the remaining fields contained valid injury data useful for analysis. City names were standardized to proper case and trimmed of leading and trailing spaces.

3.7 Missing Values Summary

Column	Missing Count	Missing %	Action Taken
Address2	96,062	91.21%	Most locations do not have a second address line - The Address 2 column was later removed due to not providing value to the analysis.
Inspection	70,013	66.48%	Not all incidents trigger a formal inspection - The inspection column was later removed due to not providing value to the analysis.
Secondary Source Title	55,380	52.58%	Not all injuries have a secondary source - The Secondary Source Title column was later removed due to not providing value to the analysis.
Latitude	70	0.07%	Retained - only 70 records, minimal impact
Longitude	70	0.07%	Retained - only 70 records, minimal impact
Address1	18	0.02%	Only 18 records, minimal impact - The Address 1 column was later removed due to not providing value to the analysis.
City	17	0.02%	Retained - only 17 records, minimal impact
Primary NAICS	2	0.00%	Retained - only 2 records, minimal impact

All missing values were retained in the columns not dropped as they represent valid data gaps rather than data quality errors. The only columns with meaningful missing data are those not used in the primary analysis; Address 1/2, Inspection, and Secondary Source Title.

3.8 NAICS Industry Name Lookup

An industry name lookup was performed using a hierarchical XLOOKUP formula in Excel and a hierarchical join in SQL and Excel against the 2022 NAICS reference table. The formula attempted matches at the 6, 4, 3, and 2 digit levels to account for codes not fully represented in the reference table. After restructuring the NAICS reference table to separate each level of the code hierarchy into individual

columns, the hierarchical lookup was able to match codes at multiple levels. This reduced the number of unmatched records to just 32, representing less than 0.03% of the total dataset. Records that could not be matched at any level were labeled as Unknown.

4. Exploratory Data Analysis

4.1 Dataset Dimensions and Structure

The dataset contains 105,317 rows and 25 columns. Each row represents one severe workplace injury incident reported to OSHA. The columns span categorical fields such as State, NatureTitle, and EventTitle, numeric outcome fields including Hospitalized, Amputation, and Loss of Eye, geographic fields including Latitude and Longitude, and reference fields including Primary NAICS and FederalState.

4.2 Unique Value Counts

Column	Unique Values	Observation
ID	105,313	Nearly unique per row — confirms minimal duplicates
Federal State	2	Only Federal or State jurisdiction — expected
Hospitalized	7	Max of 6 workers hospitalized in one incident
Amputation	3	0, 1, or 2 amputations per incident
Loss Of Eye	2	Only 0 or 1 — never more than one per incident
State	56	50 states + DC + 5 territories — all confirmed valid
Weekday	7	All 7 days represented
Direct Parent Title	40	40 broad industry categories
NAICS Industry Title	1,000	1,000 specific industries

4.3 Single Value Column Check

A single value column check confirmed that all 26 columns contained more than one unique value, indicating no columns needed to be dropped due to lack of variability. Notable findings include the Federal State column containing only two unique values representing federal and state jurisdiction respectively, and the Loss Of Eye column containing only two values confirming it as a binary outcome. The ID column returned 105,313 unique values out of 105,317 records, consistent with the single duplicate record identified and removed during data cleaning.

4.4 Statistical Summary — Outcome Variables

Metric	Hospitalized	Amputation	Loss of Eye
Total Records	105,317	105,317	105,317
Total Yes (1)	85,971	27,788	35
Total No (0)	19,346	77,522	105,277
% Yes	81.63%	26.39%	0.03%
% No	18.37%	73.61%	99.97%

A statistical summary was completed on the three primary outcome columns. Of the 105,317 records analyzed spanning January 2015 through October 2025, 81.63% resulted in hospitalization, indicating that most severe workplace injuries required hospital care. Amputations accounted for 26.39% of incidents,

meaning more than one in four severe injuries involved the loss of a limb. Loss of eye was the rarest outcome at just 0.03%, representing only 35 cases across the entire dataset.

Note: The Hospitalized and Amputation columns represent the actual count of employees impacted per incident rather than simple yes/no flags. The maximum values of 6 and 2 respectively indicate incidents where multiple workers were affected simultaneously.

4.5 Skewness Analysis

Column	Skewness	Direction	Interpretation
Hospitalized	-1.2434	Highly Negatively Skewed	Most incidents result in hospitalization - consistent with 81.63% rate
Amputation	1.0773	Highly Positively Skewed	Most incidents do not result in amputation - consistent with 26.39% rate
Loss of Eye	54.8286	Extremely Positively Skewed	Extremely rare - 99.97% of values are 0

A skewness analysis confirmed that the outcome columns are significantly non-normal, which is expected given their binary or near-binary nature. The extreme skewness of Loss of Eye (54.83) reflects the exceptionally rare nature of this outcome. These findings are consistent with the statistical summary and confirm that the dataset is heavily weighted toward hospitalization as the primary severe injury outcome.

4.6 Outlier Detection

Box plot analysis of the three outcome variables revealed that Hospitalized had 20,668 outlier records representing incidents where multiple workers were hospitalized simultaneously. Amputation showed no statistical outliers with a maximum of 2 amputations per incident. Loss of Eye produced 35 outliers consistent with the 0.03% rate identified in the statistical summary, confirming it as an extremely rare outcome.

5. Correlation Analysis

5.1 Hospitalization vs. Amputation

Using the OSHA Severe Injury Report dataset, a correlation analysis was conducted between two key injury outcome variables: Hospitalization and Amputation. The correlation coefficient was calculated at -0.79, indicating a strong negative relationship between the two variables. This means that incidents resulting in hospitalization are generally not the same incidents that result in amputation, and vice versa.

This finding is consistent with how OSHA classifies severe injury outcomes. Hospitalization and amputation are recorded as separate and largely mutually exclusive outcome flags. The strength of this correlation (-0.79) suggests this is not a coincidental pattern but rather a systematic difference in how these two injury types present. In conclusion, the negative correlation between hospitalization and amputation reflects the distinct nature of these two injury classifications within the OSHA reporting framework.

5.2 Correlation Matrix Summary

	Hospitalized	Amputation	Loss of Eye
Hospitalized	1.00	-0.79	-0.00
Amputation	-0.79	1.00	-0.01

Loss of Eye	-0.00	-0.01	1.00
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Loss of Eye showed virtually no correlation with either Hospitalization (-0.00) or Amputation (-0.01), suggesting it occurs independently of the other two outcomes and is not associated with any pattern of injury severity.

6. Key Findings

6.1 Injury Types

An analysis of the top 15 injury types revealed that Fractures were the most common severe workplace injury with 34,598 incidents, followed by Amputations at 23,448. These two categories significantly outpaced all other injury types, with the third highest category, Soreness and pain, recording only 6,522 incidents. The dominance of fractures and amputations highlights the physically dangerous nature of the industries most represented in the OSHA severe injury dataset.

6.2 Geographic Distribution

A geographic analysis of severe workplace injuries revealed that Texas recorded the highest number of incidents at 17,378, significantly outpacing all other states. Florida ranked second at 11,630, followed by Pennsylvania and Ohio at 8,314 and 8,196 respectively. The top 10 states represent a mix of large population centers, heavy manufacturing states, and states with significant construction and agricultural industries, all sectors known for elevated workplace injury risk.

6.3 Industry Analysis

An analysis of the top 20 industries by hospitalization from January 2015 through October 2025 revealed that General Medical and Surgical facilities led all industries with 2,348 hospitalizations, followed by Electrical Contractors and Commercial and Institutional Construction. Construction related industries dominated the top 20 list, appearing multiple times across different trades. Notably the healthcare industry appeared twice, in General Medical and Surgical and Nursing Care Facilities, highlighting a significant workplace injury burden in an industry dedicated to caring for others.

An analysis of broad industry categories revealed that Construction was the most dangerous sector by a significant margin, recording over 20,000 severe workplace injury outcomes, nearly three times more than the second ranked category. Manufacturing showed a notably high proportion of amputations relative to hospitalizations, suggesting the intensive nature of that sector with the use of machinery carries elevated amputation risk.

6.4 Day of Week Analysis

An analysis of severe workplace injuries by day of week from January 2015 through October 2025 revealed that Tuesday had the highest number of incidents at 19,660, followed closely by Wednesday at 19,718 and Thursday at 18,894. Mid-week days consistently showed the highest injury counts, which may reflect peak operational activity during those days. Friday showed a notable decline at 16,305 and weekend days were significantly lower with Saturday at 7,514 and Sunday at 4,736, consistent with reduced workforce activity on non-traditional workdays.

6.5 Year Over Year Trend

A year over year analysis of severe workplace injuries from 2015 through October 2025 revealed a clear upward trend from 2015 to 2018, peaking at 11,156 incidents. A sharp decline occurred in 2020

coinciding with the COVID-19 pandemic, dropping to 8,915, a reduction of over 20% from the 2018 peak. Incidents continued to decline in 2021 before showing a partial recovery in 2022. Notably severe injury counts have not returned to pre-pandemic levels, suggesting possible lasting changes in workplace safety practices or workforce activity patterns. The 2025 figure represents only a partial year through October and is not directly comparable to prior full year totals.

6.6 Forecasting

A time series forecast was generated using linear trend analysis in Python and exponential smoothing in Tableau to project severe workplace injury incidents through 2026-2027. Both models suggest monthly incidents will stabilize between 700 and 800 reports per month. The Tableau forecast confidence interval widens progressively into the future reflecting increasing uncertainty over longer forecast horizons. Notably the model does not project a return to pre-pandemic incident levels, suggesting the post-COVID reduction in severe workplace injuries may represent a structural shift rather than a temporary decline.

7. SQL Analysis

7.1 NAICS Join for Industry Analysis

The NAICS and severe_injuries tables are joined together to enrich the injury data with human-readable industry names based on the NAICS code assigned to each incident. The NAICS table was referenced four times using different aliases (n6, n4, n3, and n2), with each one representing a different matching strategy. The matching starts by looking for an exact 6-digit match. If that does not find a result, it moves to 4 digits, then 3, and then 2. The decision was made to skip the 5-digit match to avoid duplication and redundancy, which was confirmed through analysis completed in Microsoft Excel.

The LEFT JOINs are what build the foundation for this strategy. Each LEFT JOIN creates what can be thought of as a phantom column in the background, lining up all four match attempts side by side for each injury record. The aliases n6, n4, n3, and n2 label each of those phantom columns so the query knows which match attempt it is referring to. This temporary structure only exists for the life of the query and disappears the moment the results are returned.

The COALESCE function then takes over and controls the hierarchy. It reads across those phantom columns from left to right and returns the first value it finds that is not empty. If none of the match attempts return a result, it defaults to 'unknown industry.' Neither function could accomplish this alone; the LEFT JOINs build the phantom columns and COALESCE decides which one to use.

LEFT JOIN was the right choice for this analysis because it keeps every record from the severe_injuries table regardless of whether a match was found. A standard INNER JOIN or INTERSECT would have dropped any record without an exact match, resulting in a significant loss of data. A RIGHT JOIN was also ruled out because it would have prioritized keeping every record from the NAICS reference table instead of the injury data, which was the opposite of what was needed.

7.2 Match Status Analysis

This query was written to validate how well the NAICS join performed by counting how many injury records successfully matched to an industry name versus how many came back as unknown. The CASE statement simplifies the result down to just two categories; 'matched' or 'unknown industry.' It does this by checking whether the COALESCE function returned a value or came back empty. The COUNT() function then totals up how many records fell into each category, and GROUP BY match_status splits those totals into two separate rows in the results.

7.3 Employer Name Standardization

Employer names were standardized in SQL using a combination of TRIM to remove leading and trailing spaces, COALESCE to handle null and blank values, and a CASE statement to standardize common employer name variations. The CASE statement identified frequently occurring employers with multiple name formats, such as Walmart, UPS, Home Depot, and FedEx, and consolidated them into a single standardized name. This approach reduced the number of unique employer entries and improved the accuracy of employer-level analysis.

8. Python Analysis

8.1 Libraries Used

The following Python libraries were used throughout the analysis: pandas for data loading, cleaning, and manipulation; numpy for numerical calculations including linear trend analysis; matplotlib for chart creation and formatting; and seaborn for chart styling.

8.2 Data Cleaning in Python

Upon loading the CSV file into Python, several data type corrections were required. The EventDate column was stored as an Excel serial number and required conversion using `pd.to_datetime()`. The Zip column was converted to text and zero-padded to 5 digits using `str.zfill(5)`. The Amputation and Loss of Eye columns contained float values due to blank entries and were converted to integers after filling nulls with zero. Employer names were standardized using `str.strip()`, `str.title()`, and regex-based space normalization.

8.3 Charts Produced

Chart	Type	Key Finding
Outcomes Distribution	Pie Chart	81.63% hospitalized, 26.39% amputation, 0.03% loss of eye
Injuries by Nature Title	Horizontal Bar	Fractures (34,598) and Amputations (23,448) dominate
Top 10 States	Horizontal Bar	Texas leads with 17,378 incidents
Correlation Heatmap	Heatmap	Strong negative correlation (-0.79) between hospitalization and amputation
Incidents Over Time	Line Chart	Peak 2018-2019, COVID dip 2020, no full recovery
Day of Week	Bar Chart	Tuesday and Wednesday peak, weekends significantly lower
Top 20 Industries	Stacked Bar	General Medical leads, construction trades dominate
Top 20 Broad Categories	Stacked Bar	Construction nearly 3x second place
Forecast	Line Chart	Monthly incidents projected to stabilize 700-800
Outlier Detection	Box Plot	20,668 hospitalization outliers, 35 loss of eye outliers

9. Tableau Analysis

9.1 Data Connection

Tableau was connected to the processed CSV file `severe_injuries_processed_v2.csv` which contained all cleaned data, industry name lookups, and calculated fields from the Python analysis. No additional joins were required in Tableau as all necessary columns were already present in the processed file.

9.2 Data Cleaning in Tableau

Upon connecting the data source, it was discovered that the Direct Parent Title column still contained trailing 'T' characters on several industry category names — for example 'Construction' displayed correctly but others such as 'Machinery ManufacturingT' and 'Chemical ManufacturingT' retained the formatting artifact from the original NAICS reference table. To resolve this a calculated field named Direct Parent Title Clean was created using the following formula:

```
IF RIGHT([Direct Parent Title], 1) = "T"
THEN LEFT([Direct Parent Title], LEN([Direct Parent Title]) - 1)
ELSE [Direct Parent Title]
END
```

This calculated field was used in place of the original Direct Parent Title column throughout all worksheets and dashboards to ensure clean industry category names were displayed consistently across all visualizations.

9.3 Worksheets Built

Worksheet	Chart Type	Key Finding
OSHA Severe Injury Outcomes by State	Bar Chart	Texas leads all states with interactive year filter
Severe Injury Outcomes by Direct Parent Title	Bar Chart	Construction dominates all broad industry categories
Monthly Reports Over Time	Line Chart	COVID-19 impact clearly visible with trend line
Forecasted Reports	Line Chart	Incidents projected to stabilize 700-800 per month
Reports by Day of Week	Bar Chart	Mid-week peak with weekend contrast clearly visible
Severe Injury Report by Broad Category	Bubble Chart	Construction bubble dominates visual representation
Hospitalized Pie	Pie Chart	81.63% hospitalized vs 18.37% not hospitalized
Amputation Pie	Pie Chart	26.39% amputation vs 73.61% no amputation
Loss of Eye Pie	Pie Chart	0.03% loss of eye vs 99.97% no loss of eye
Top 20 Injuries by Nature Title	Horizontal Bar	Fractures lead at 34,598 followed by Amputations at 23,449
Top 20 Injuries by Body Part	Horizontal Bar	Fingertips lead at 10,402 — fingers dominate top 3
Total Reports KPI	Text/Number	105,318 total severe injury reports

9.4 Dashboards

Three interactive dashboards were created to present the analysis in a clean and focused format.

Dashboard 1 - Injury Overview

Combined a Total Reports KPI card showing 105,318 total records, three pie charts showing Yes vs No outcomes for Hospitalized, Amputation, and Loss of Eye, and two horizontal bar charts showing Top 20 Injuries by Nature Title and Top 20 Injuries by Body Part. A Year filter was applied across all worksheets allowing users to explore how injury types and body parts affected changed year over year across the full reporting period from January 2015 through October 2025.

Dashboard 2 - Time Analysis

Combined the Monthly Reports Over Time, Forecasted Reports, and Reports by Day of Week worksheets. A Year filter was applied across all worksheets to allow interactive exploration by year. The COVID-19 reference line is visible on both time series charts highlighting the sharp decline in reported severe injuries beginning in early 2020.

Dashboard 2 - Industry Analysis

Combined the OSHA Severe Injury Outcomes by State, Severe Injury Outcomes by Direct Parent Title, and Severe Injury Report by Broad Category worksheets. Year and Direct Parent Title Clean filters were applied to allow interactive filtering across all three visualizations. The Direct Parent Title Clean calculated field was used throughout this dashboard to ensure trailing T artifacts were removed from all industry category labels.

10. Narrative

10.1 Target Audience

This analysis is intended for two primary audiences. The first is workplace safety professionals, including OSHA compliance officers, safety managers, and risk assessment teams, who are responsible for identifying injury trends, evaluating program effectiveness, and implementing corrective actions across their organizations. The second is industry HR leaders who are responsible for workforce wellbeing, employee relations, return-to-work programs, and compliance reporting. Both audiences benefit from understanding which industries, injury types, and time periods carry the highest severe injury risk so they can make informed, data-driven decisions about where to focus safety resources.

10.2 Questions Intended to Answer

This analysis was designed to answer the following questions:

1. Which industries have the highest rates of severe workplace injuries including hospitalization and amputation?
2. How has the frequency of severe workplace injuries changed over time from 2015 through 2025?
3. Which injury types and body parts are most affected in severe workplace incidents?
4. Is there a relationship between the type of outcome, hospitalization versus amputation, and the industry or event type?
5. Are there patterns in when severe injuries occur, such as certain days of the week or months of the year?
6. What does the forecast suggest about the future trajectory of severe workplace injuries?

10.3 Steps Taken to Answer These Questions

The analysis followed a structured multi-tool approach. Data cleaning was performed in Excel and SQL to standardize column formats, handle missing values, remove duplicates, and enrich the dataset with

industry names through a hierarchical NAICS code lookup. Exploratory data analysis was conducted in Python using descriptive statistics, distribution charts, correlation analysis, and outlier detection. Trend analysis and forecasting were performed using both Python and Tableau. Interactive dashboards were built in Tableau to allow stakeholders to filter findings by year and industry category.

10.4 Assumptions Made

Several assumptions were made during this analysis. First, it was initially assumed that the Hospitalized, Amputation, and Loss of Eye columns were binary 0/1 flags. Upon further investigation it was confirmed that Hospitalized and Amputation represent actual counts of affected employees per incident, with maximum values of 6 and 2 respectively. This assumption was corrected and the analysis updated accordingly.

Second, it was assumed that NAICS codes in the OSHA dataset could be matched to the 2022 NAICS reference table using a hierarchical fallback approach. This assumption proved reasonable, a 97.48% match rate was achieved, though a small number of codes representing older or non-standard classifications could not be matched at any level and were labeled as Unknown.

Third, it was assumed that the COVID-19 pandemic was the primary driver of the sharp decline in reported severe injuries observed in 2020. While this is consistent with the broader pattern of reduced workplace activity during that period, it cannot be confirmed with certainty from this dataset alone as reporting behavior may also have changed during that time.

10.5 Industry Standard Metrics and Measurements

This analysis references OSHA's mandatory severe injury reporting requirements, which define three reportable severe injury outcomes: hospitalization within 24 hours of a work-related incident, amputation, and loss of an eye. These definitions are codified in OSHA regulation 29 CFR 1904.39 and serve as the basis for the three primary outcome variables analyzed throughout this project. The NAICS classification system maintained by the U.S. Census Bureau was used as the industry standard framework for categorizing employers by industry type.

10.6 Limitations

Several limitations should be noted when interpreting this analysis. First, the dataset relies on employer-submitted reports, which means underreporting is possible. Employers may not always report every qualifying incident, and reporting practices may vary by industry, company size, and jurisdiction. Second, employer name standardization was limited by the inconsistency of manually entered names in the OSHA system. While common employer variations were standardized, many variations likely remain due to the volume and diversity of entries. Third, 2.52% of records could not be matched to an industry name even after applying the hierarchical NAICS lookup strategy, which may slightly underrepresent certain industries in the industry-level analysis. Fourth, the 2025 data covers January through October and is therefore not directly comparable to prior full year totals. Finally, the correlation and trend analyses identify patterns in the data but do not establish causation, additional research would be needed to confirm the underlying reasons for observed trends.

10.7 Recommendations

Based on the findings of this analysis the following recommendations are offered to workplace safety professionals and industry HR leaders:

1. Root Cause Analysis in High-Risk Industries: The industries with the highest severe injury rates, particularly Construction, General Medical and Surgical Hospitals, Electrical Contractors, and Manufacturing, should conduct targeted root cause analyses to identify the specific conditions, tasks, and

environments contributing to their elevated injury rates. Understanding the why behind the numbers is critical to designing effective interventions.

2. Targeted Safety Training: Organizations in the top 20 industries by hospitalization and amputation should develop and implement more targeted safety training programs specific to their most common injury types. For example, construction organizations should focus on fracture prevention and fall protection, while manufacturing organizations should prioritize machine guarding and amputation prevention given their disproportionately high amputation rates.

3. Mid-Week Safety Focus: Given that Tuesday, Wednesday, and Thursday consistently show the highest injury counts, organizations should consider scheduling additional safety checks, toolbox talks, and supervisor observations during mid-week peak activity periods.

4. Sustain Post-COVID Safety Gains: The decline in severe injuries following 2020 has not fully reversed, suggesting that some post-pandemic changes in workplace practices may have had a positive safety impact. Organizations should identify and preserve any changes made during that period, such as reduced workforce density, increased remote supervision, or enhanced PPE protocols, that may have contributed to lower injury rates.

5. State-Level Focus: Texas, Florida, Pennsylvania, and Ohio consistently rank among the highest injury states and should be prioritized for state-level safety initiatives, industry partnerships, and targeted OSHA outreach efforts.

10.8 Presentation Approach

During the presentation this analysis will be explained by walking through each tool in sequence, Excel for data cleaning and initial exploration, SQL for database querying and industry joins, Python for statistical analysis and visualizations, and Tableau for interactive dashboards. Each section will highlight not just what was found but why specific methods were chosen, including the rationale for the hierarchical NAICS matching strategy, the decision to use LEFT JOIN over other join types, and the reasoning behind retaining rather than removing missing value records. The Tableau dashboards will be demonstrated live to show how stakeholders can interact with the data to answer their own questions.

11. Conclusion

This capstone project analyzed 105,317 OSHA severe workplace injury reports spanning January 2015 through October 2025 using four analytical tools, Microsoft Excel, MySQL/DBBeaver, Python/Jupyter Notebook, and Tableau.

The analysis revealed that the Construction industry carries the highest severe injury burden across all broad industry categories, followed by Manufacturing and Healthcare. Fractures and Amputations are the most common injury types, and Texas, Florida, Pennsylvania, and Ohio consistently rank among the highest injury states.

A significant decline in severe workplace injuries was observed beginning in 2020 coinciding with the COVID-19 pandemic. Incidents have not returned to pre-pandemic levels, and forecasting models suggest monthly incidents will stabilize between 700 and 800 reports per month through 2027.

Mid-weekdays, particularly Tuesday and Wednesday, consistently show the highest injury counts, suggesting that peak operational activity during mid-week may contribute to elevated injury risk.

The strong negative correlation (-0.79) between hospitalization and amputation outcomes confirms that these injury types are largely mutually exclusive, reflecting distinct mechanisms of injury and reinforcing the importance of analyzing each outcome variable independently.

This analysis provides actionable insights for workplace safety professionals, policymakers, and industry stakeholders seeking to understand the patterns and trends in severe workplace injuries across the United States.